D.4.1 User Requirements and Analysis Document

PlanER: Cost-Based Optimization of Hybrid Classical–LLM Entity Resolution Plans in a Document Database

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1. Introduction & Scope

1.1 Purpose

This document defines the functional and non-functional requirements for **PlanER**, a system that treats entity resolution (ER) as a declarative request compiled, through a cost-based optimizer, into a physical execution plan over MongoDB. It is the agreed reference between the project team and the course evaluators for what the system must do, how well it must do it, and how each requirement will be verified. Every requirement is derived from the D.2.1 Full Project Proposal.

1.2 Objectives

- Let a user submit one declarative ER request (two collections, an attribute mapping, and a quality target θ or a budget B) instead of hand-coding a matching pipeline.
- Automatically choose a physical plan (blocking access path, ordered matcher cascade, confidence thresholds, clustering method) that meets the objective at lowest cost or highest achievable quality.
- Execute the plan natively in MongoDB as aggregation pipelines over B-tree, multikey, full-text (**$search**) and vector (**$vectorSearch**) indexes.
- Show, on five public ER benchmarks, where non-LLM matchers dominate the cost–accuracy frontier and how much of an LLM's accuracy a cheap cascade retains.

1.3 Project Boundaries

In scope: declarative request handling; four blocking access paths; a five-tier matcher cascade (rules, Fellegi–Sunter, random forest, local LLM, hosted LLM); the cost-based optimizer (sampling, estimation, Pareto-pruned plan enumeration); execution and run logging; clustering with transitivity repair; a Streamlit demo; evaluation against LLM-only, classical-only and embedding-proxy baselines on DBLP–ACM, Abt–Buy, Amazon–Google, DBLP–Scholar and (stretch) WDC Products.
Out of scope: production or multi-tenant deployment; a general query language beyond the fixed ER request; streaming re-resolution; engines other than MongoDB; enterprise authentication, auditing or role-based access control; a polished GUI beyond the demo.

1.4 Stakeholder Roles

Stakeholder	Role	Responsibilities
Project team (S. Patel, N. Syed)	Developers / researchers	Design and implement blocking, matchers, optimizer and demo; run experiments; write the final report. One leads blocking & storage, the other matching & optimization; both contribute to every phase.
Course instructor & TAs	Evaluators	Review deliverables against the rubric, approve milestones, grade report, demo and presentation.
Data engineer / analyst (user persona)	System user	Submits ER requests with a budget or quality target; interprets the returned plan, clusters and cost/F1 report; adjusts the objective and re-runs.
Database administrator (held by the team)	Infrastructure owner	Deploys and maintains MongoDB 8.2 and the mongot search process in Docker; manages indexes and storage.
Domain labeler (benchmark ground truth)	Quality provider	Supplies the labeled sample the optimizer uses to estimate matcher cost, escalation rate and accuracy.

2. Functional Requirements

Prioritized with MoSCoW (**M** = Must, **S** = Should, **C** = Could). "Shall" denotes a mandatory, testable behavior.

#	Requirement	Pri.
FR-1	The system shall accept a declarative ER request: two collections (or one for deduplication), an attribute mapping, and an objective — a quality target θ (minimize cost s.t. F1 ≥ θ) or a budget B (maximize F1 s.t. cost ≤ B).	M
FR-2	The system shall load records into MongoDB and derive a normalized text field, token array, blocking key and 384-dimensional sentence embedding per record.	M

#	Requirement	Pri.
FR-3	The system shall generate candidate pairs with a B-tree index and **$lookup** on the blocking key.	M
FR-4	The system shall generate candidate pairs with a multikey index over the token array (shared rare tokens).	M
FR-5	The system shall generate candidate pairs with a full-text index via **$search** (BM25 top-k).	S
FR-6	The system shall generate candidate pairs with a vector index via **$vectorSearch** (HNSW top-k over embeddings).	M
FR-7	The system shall score candidate pairs with a string-similarity rule matcher.	M
FR-8	The system shall score candidate pairs with an unsupervised Fellegi–Sunter matcher fitted by expectation–maximization.	M
FR-9	The system shall score candidate pairs with a random-forest matcher trained on similarity features.	S
FR-10	The system shall escalate pairs a matcher cannot confidently decide to the next, costlier matcher, up to a local LLM served through Ollama.	M
FR-11	The system shall optionally escalate the most uncertain residual pairs to one hosted LLM to obtain real per-token cost.	C
FR-12	The system shall cache every LLM verdict in a dedicated collection keyed by a hash of the pair, and reuse it instead of re-invoking the LLM.	M
FR-13	The system shall draw a stratified labeled sample and use it to estimate each matcher's per-pair cost, escalation rate, precision and recall, and each access path's candidate count, latency and pair completeness.	M
FR-14	The system shall enumerate plans (access path × matcher subset/order × threshold grid) and prune plans dominated in both estimated cost and F1.	M
FR-15	The system shall select from the Pareto set the plan that minimizes cost for the quality target, or maximizes F1 within the budget.	M
FR-16	The system shall execute the selected plan as MongoDB aggregation pipelines and record explain() output for every blocking stage.	M
FR-17	The system shall form entity clusters from matched pairs using connected components.	M
FR-18	The system shall detect transitivity violations (A–B and B–C matched, A–C not) and repair them without additional LLM calls.	S
FR-19	The system shall log, per executed plan, observed dollar cost, compute time, LLM calls, pairwise and cluster F1, and labeling cost in a run_log collection.	M
FR-20	The system shall report optimizer prediction error by comparing estimated against observed cost and F1.	M
FR-21	The system shall provide a Streamlit interface to submit a request, view the chosen plan, move the target F1 or budget, and watch cost and F1 update.	S
FR-22	The system shall evaluate each selected plan against an LLM-only pipeline, the best single classical matcher and a LOTUS-style embedding-proxy cascade on the same dataset.	S
FR-23	The system shall support all five benchmark dataset pairs as interchangeable inputs through configuration only.	C

3. Non-Functional Requirements

#	Category	Requirement
NFR-1	Performance	Sample-based estimation (FR-13) shall finish in under 5 minutes for a stratified sample of a few hundred pairs on development hardware.
NFR-2	Scalability	Blocking shall reduce the DBLP–Scholar workload (~168 million possible pairs) to a candidate set processed without exhaustive comparison, for every access path in FR-3 to FR-6.
NFR-3	Reliability	The verdict cache (FR-12) shall ensure an identical repeat run issues zero new LLM calls (100% cache hits).
NFR-4	Data integrity	Every match or cluster document shall reference record identifiers that exist in the source collections at run time.
NFR-5	Resilience	Because **$search**/**$vectorSearch** (mongot) are public-preview features, text and vector paths shall sit behind a common interface with a fallback to the free MongoDB Atlas tier.
NFR-6	Security	API keys and MongoDB credentials shall come from environment variables or a local secrets file — never committed to source control or written to run_log.

#	Category	Requirement
NFR-7	Reproducibility	Local LLMs shall use greedy decoding and a fixed prompt; training, estimation and test splits shall be disjoint to prevent label leakage.
NFR-8	Usability	A user without MongoDB or ML background shall read the plan, cost and F1 in the demo without consulting source code.
NFR-9	Maintainability	Blocking, matchers, optimizer and clustering shall be separate Python modules with documented interfaces.
NFR-10	Portability	The full stack (MongoDB 8.2, mongot, PyMongo, Ollama) shall run in Docker so both members' environments are identical.
NFR-11	Cost compliance	For a budget B, observed plan cost shall not exceed B except in logged overruns attributed to estimation error.

4. Acceptance Criteria

#	Verifies	Measurable acceptance criterion
AC-1	FR-1, FR-15	For θ = 0.85, the returned plan's **observed** F1 is within 0.05 of θ on at least 4 of 5 datasets.
AC-2	FR-3 – FR-6	Each blocking path reaches ≥ 90% pair completeness on DBLP–ACM and Abt–Buy; explain() output is logged for every run.
AC-3	FR-7 – FR-10	The cascade cuts LLM calls by ≥ 70% versus the LLM-only baseline while pairwise F1 drops by ≤ 5 points.
AC-4	FR-13, FR-20	Predicted cost is within 15% and predicted F1 within 0.05 of observed values on at least 3 of 5 datasets.
AC-5	FR-12, NFR-3	Re-submitting an identical request on unchanged data causes zero new LLM calls (logged 100% cache-hit rate).
AC-6	FR-17, FR-18	On a dataset with ≥ 20 transitivity violations, repair reduces them by a logged percentage with zero extra LLM calls.
AC-7	FR-15, NFR-11	Across 20 trial runs with budget B, observed cost stays ≤ B in at least 95% of runs; exceedances are logged with cause.
AC-8	FR-21	In the demo, a submitted request shows the chosen plan, live cost and live F1 within 5 seconds (sample-based estimate).
AC-9	FR-22	Each dataset's report lists four comparable results (PlanER, LLM-only, best classical, embedding-proxy) on the same held-out test split.
AC-10	NFR-2	On DBLP–Scholar, blocking completes and reports pair completeness and reduction ratio without an exhaustive 168M-pair comparison.

5. Revision History

Version	Date	Author(s)	Description of changes	Status
v0.1	2026-10-05	S. Patel, N. Syed	Initial draft from the D.1 Pre-Proposal and D.2.1 Full Proposal; scope, stakeholders, initial requirements.	Draft
v0.2	2026-10-05	S. Patel, N. Syed	Expanded functional requirements (all access paths, five-tier cascade), added MoSCoW priorities and NFRs.	Draft
v1.0	2026-10-05	S. Patel, N. Syed	Added measurable acceptance criteria mapped to requirements; finalized as D.4.1.	Submitted
v1.1	pending	pending	Reserved for instructor/TA feedback and resulting revisions.	Pending approval